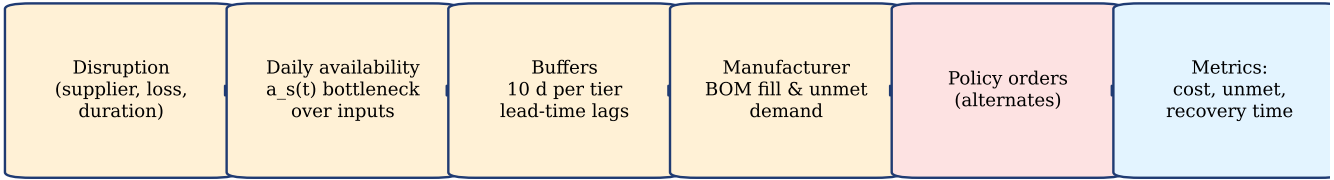Cascading-disruption simulation